CLASS 8 · PART TEST-1 · STUDENT PERFORMANCE REPORT

Madipadiga Dheeraj

Pragya G8 (EXI) Achievers · Adm No **257381393**

Campus HYD/Nallakunta DD Colony
JEE Mains 08 Oct 2026
JEE Advanced 09 Oct 2026

JEE Mains

08 Oct 2026 · 74 questions

147 / 296 — **49.7%**

184	128	10/63	57
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **178.8 / 300** (59.6%) → this test **49.7%** ▼ **9.9**

149 marks lost: 125 on 25 wrong answers (incl. 25 negative) · 24 on 6 skipped

JEE Advanced

09 Oct 2026 · 48 questions

69 / 171 — **40.4%**

431	301	20/63	123†
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **99.0 / 180** (55.0%) → this test **40.4%** ▼ **14.6**

102 marks lost: 95 on 24 wrong answers (incl. 12 negative) · 7 on 2 skipped

1 Subject scorecard

Your marks in each subject, against the 63 students of this program on campus and against your own weekly-test average.

SUBJECT	MARKS	SCORE VS CAMPUS	SUBJECT AIR	CAMPUS AVG / BEST	WEEKLY AVG	CHANGE	RIGHT / WRONG / SKIP
JEE Mains 147 / 296 · 49.7%							
Maths	51/96	53%	121	37.0 / 77	67.7/100 67.7%	▼ 14.6	15 / 9 / 0
Physics	30/100	30%	333	18.3 / 60	63.2/100 63.2%	▼ 33.2	10 / 10 / 5
Chemistry	66/100	66%	308	37.4 / 86	47.8/100 47.8%	▲ 18.2	18 / 6 / 1
JEE Advanced 69 / 171 · 40.4%							
Maths	28/54	52%	223	24.2 / 41	32.5/60 54.2%	▼ 2.4	8 / 7 / 0
Physics	4/60	7%	1801	9.4 / 40	34.8/60 58.0%	▼ 51.4	3 / 13 / 1
Chemistry	37/57	65%	339	23.9 / 50	31.5/60 52.5%	▲ 12.4	11 / 4 / 1

Bar = your score % | campus average | best on campus Change = this test % − weekly-test average %

What this report says

1. **Chemistry** is your strongest JEE Mains subject: 66/100 (66%), All-India subject rank 308.
2. **Physics** needs the most work in JEE Mains: 30/100 (30%), subject rank 333; you lost 15 of its 25 questions.
3. **Chemistry** is your strongest JEE Advanced subject: 37/57 (65%), All-India subject rank 339.
4. **Physics** needs the most work in JEE Advanced: 4/60 (7%), subject rank 1801; you lost 14 of its 17 questions.
5. Biggest fall against your weekly-test average: **JEE Advanced Physics** – 7% here against 58% on average.
6. Biggest improvement: **JEE Mains Chemistry** – 66% here against 48% on average.
7. **14 questions** you lost were answered correctly by 70%+ of the batch's top 100 – about **66 marks** within easy reach (Mains Q18, Q26, Q28, Q32, Q61, Q70, Q74 · Advanced Q3, Q19, Q24, Q25, Q38, Q40, Q47).
8. Negative marking cost you **25 in JEE Mains and 12 in JEE Advanced**.
9. You attempted almost everything (only 8 questions skipped across both papers), so the marks to gain are in **accuracy**: wrong answers – 25 in JEE Mains and 24 in JEE Advanced.

JEE Mains – question by question

147 / 296 · 49.7%
08 Oct 2026

2 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Physics	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
Chemistry	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75

3 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 63 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40-69% · **Tough one** below 40%

Maths

9 of 24 questions lost · 45 marks · 1 easy miss · 5 within reach · 3 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q18	Easy miss	**Surds & Indices**	Single correct · +4 / −1	Wrong	5	75%	67%
Q13	Within reach	**Quadratic Equations**	Single correct · +4 / −1	Wrong	5	63%	27%
Q25	Within reach	**Surds & Indices**	Numerical · +4 / −1	Wrong	5	44%	21%
Q3	Within reach	**Sequences & Series** – Product of GP terms; natural-number GP	Single correct · +4 / −1	Wrong	5	42%	51%
Q4	Within reach	**Sequences & Series** – Identify AP / GP / HP from a surd expression	Single correct · +4 / −1	Wrong	5	42%	19%
Q24	Within reach	**Logarithms & Exponents**	Numerical · +4 / −1	Wrong	5	40%	78%
Q23	Tough one	**Surds & Indices** – Cube root of a surd expression	Numerical · +4 / −1	Wrong	5	24%	13%
Q5	Tough one	**Quadratic Equations** – Range of a rational quadratic expression	Single correct · +4 / −1	Wrong	5	21%	6%
Q16	Tough one	**Quadratic Equations** – Roots in GP, condition on discriminant	Single correct · +4 / −1	Wrong	5	19%	3%

Physics

15 of 25 questions lost · **70** marks · 3 easy miss · 5 within reach · 7 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q26	Easy miss	**Vectors**	Single correct · +4 / −1	Wrong	5	91%	86%
Q28	Easy miss	**Vectors** – Resultant ⊥ smaller vector; find magnitudes	Single correct · +4 / −1	Wrong	5	77%	30%
Q32	Easy miss	**Kinematics**	Single correct · +4 / −1	Wrong	5	75%	29%
Q29	Within reach	**Vectors** – Angles between three vectors summing to zero	Single correct · +4 / −1	Wrong	5	62%	24%
Q40	Within reach	**Laws of Motion** – Velocity over fixed distance vs mass ($v \propto 1/\sqrt{m}$)	Single correct · +4 / −1	Wrong	5	62%	30%
Q44	Within reach	**Laws of Motion** – Normal reactions on a cylinder resting on edges	Single correct · +4 / −1	Wrong	5	53%	29%
Q30	Within reach	**Laws of Motion** – Tension / normal reaction in two-block systems (matrix match)	Single correct · +4 / −1	Wrong	5	42%	30%
Q35	Within reach	**Kinematics** – $a = kv^2$ – integrate to get v(s)	Single correct · +4 / −1	Skipped	4	41%	5%
Q38	Tough one	**Kinematics** – Resistance $\propto v^2$ – time inside plank	Single correct · +4 / −1	Skipped	4	27%	11%
Q45	Tough one	**Laws of Motion** – Contact force between pushed blocks (statement type)	Single correct · +4 / −1	Wrong	5	27%	10%
Q36	Tough one	**Kinematics** – a(t) with modulus – area under a-t	Single correct · +4 / −1	Wrong	5	24%	5%
Q43	Tough one	**Laws of Motion** – Normal reaction across an inclined contact	Single correct · +4 / −1	Wrong	5	20%	10%
Q47	Tough one	**Vectors** – Net displacement from multi-leg journey	Numerical · +4 / −1	Skipped	4	15%	3%
Q48	Tough one	**Kinematics** – Distance from a-t graph	Numerical · +4 / −1	Skipped	4	15%	3%
Q49	Tough one	**Kinematics** – Free fall then deceleration – find height	Numerical · +4 / −1	Skipped	4	15%	2%

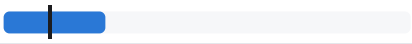
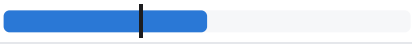
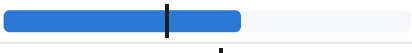
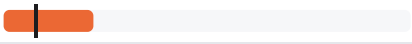
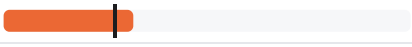
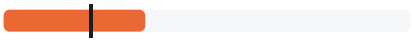
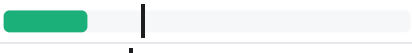
Chemistry

7 of 25 questions lost · **34** marks · 3 easy miss · 3 within reach · 1 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q61	Easy miss	**Atomic Structure**	Single correct · +4 / −1	Wrong	5	84%	40%
Q70	Easy miss	**Chemical Bonding**	Single correct · +4 / −1	Wrong	5	72%	52%
Q74	Easy miss	**Periodic Classification**	Numerical · +4 / −1	Wrong	5	70%	46%
Q53	Within reach	**Mole Concept & Basics** – Valence electrons in N^{3-} (mole → electrons)	Single correct · +4 / −1	Wrong	5	65%	25%
Q75	Within reach	**Chemical Bonding** – Hybrid orbitals + lone pairs in NH_3	Numerical · +4 / −1	Skipped	4	60%	30%
Q59	Within reach	**Chemical Bonding** – VBT, orbital overlap, hybridisation statements	Single correct · +4 / −1	Wrong	5	50%	29%
Q67	Tough one	**Chemical Bonding** – Strongest ionic bond (charge and size)	Single correct · +4 / −1	Wrong	5	32%	13%

4 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Quadratic Equations	5	**5**/20		11%	Q5, Q13, Q16
Sequences & Series	5	**10**/20		33%	Q3, Q4
Logarithms & Exponents	3	**7**/12		40%	Q24
Surds & Indices	11	**29**/44		53%	Q18, Q23, Q25
Physics					
Laws of Motion	8	**7**/32		7%	Q30, Q40, Q43, Q44, Q45
Vectors	7	**9**/28		27%	Q26, Q28, Q29, Q47
Kinematics	10	**14**/40		21%	Q32, Q35, Q36, Q38, Q48, Q49
Chemistry					
Chemical Bonding	6	**5**/24		34%	Q59, Q67, Q70, Q75
Atomic Structure	5	**15**/20		31%	Q61
Periodic Classification	5	**15**/20		58%	Q74
Mole Concept & Basics	9	**31**/36		32%	Q53

Bar = your % of the chapter's marks | campus average

5 JEE Mains · Maths – questions to revisit

9 questions · easy misses first
cropped from the question paper

Q18 Surds & Indices

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **75%** got it right · Campus: **67%** (42 of 63)

Easy miss
You: Wrong

18. Statement-1: $\sqrt{6-\sqrt{7}+\sqrt{27+4\sqrt{35}}}=1+\sqrt{5}$

Statement-2: If $x=\sqrt{7}+\sqrt{3}, y=\sqrt{7}-\sqrt{3}, xy=4$ then $x^4+y^4=352$

(A)Statement 1is FALSE and Statement 2 is FALSE.
(B)Statement 1is TRUE and Statement 2 is TRUE.
(C)Statement 1is TRUE and Statement 2 is FALSE.
(D)Statement 1 is FALSE but Statement 2 is TRUE.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q13 Quadratic Equations

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **63%** got it right · Campus: **27%** (17 of 63)

Within reach
You: Wrong

13. The value of α for which the sum of the squares of the roots of equation $x^2-(\alpha-2)x-\alpha-1=0$ assume the least value, is

(A)1 (B)0 (C)3 (D)2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q25 Surds & Indices

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5
Top 100: **44%** got it right · Campus: **21%** (13 of 63)

Within reach
You: Wrong

25. If $\left(2+\sqrt{3}\right)^{x^2-7x+1}+\left(2-\sqrt{3}\right)^{x^2-7x-1}=\frac{2}{2-\sqrt{3}}$, then the sum of values of 'x' satisfying this equation is equal to_____ ?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q3

Sequences & Series – Product of GP terms; natural-number GP

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **42%** got it right · Campus: **51%** (32 of 63)

Within reach

You: Wrong

3. If $x_1, x_2, x_3, x_4, x_5, \ldots$ is a geometric progression of natural numbers and $x_1x_2x_3x_4 = 64$, then complete set of solutions of x_4 is

(A) $\{1\}$ (B) $\{8\}$ (C) $\{1, 8\}$ (D) cannot be determined

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q4

Sequences & Series – Identify AP / GP / HP from a surd expression

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **42%** got it right · Campus: **19%** (12 of 63)

Within reach

You: Wrong

4. If a, b, c are in A.P., then $\frac{1}{\sqrt{a}+\sqrt{b}}, \frac{1}{\sqrt{a}+\sqrt{c}}, \frac{1}{\sqrt{b}+\sqrt{c}}$ are in

(A) A.P. (B) G.P. (C) H.P. (D) Neither A.P nor G.P nor H.P

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q24

Logarithms & Exponents

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5

Top 100: **40%** got it right · Campus: **78%** (49 of 63)

Within reach

You: Wrong

24. The value of $6+\log_{\frac{3}{2}}\left(\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\ldots\ldots}}}\right)$ is___?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q23 — Surds & Indices – Cube root of a surd expression

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5

Top 100: **24%** got it right · Campus: **13%** (8 of 63)

Tough one

You: Wrong

23. If $\sqrt[3]{38\sqrt{14}-100\sqrt{2}}=\sqrt{x}-\sqrt{y}$, then the value of $x-y$ is__?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q5 — Quadratic Equations – Range of a rational quadratic expression

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **21%** got it right · Campus: **6%** (4 of 63)

Tough one

You: Wrong

5. If x is real, then $\dfrac{x^2+2x+c}{x^2+4x+3c}$ can take all real values if

(A) $0<c<2$ (B) $0\le c\le 1$

(C) $-1<c<1$ (D) $0<c<1$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q16 — Quadratic Equations – Roots in GP, condition on discriminant

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **19%** got it right · Campus: **3%** (2 of 63)

Tough one

You: Wrong

16. Let α, β be the roots of the equation $ax^2+bx+c=0$ and $\Delta=b^2-4ac$. If $\alpha+\beta, \alpha^2+\beta^2, \alpha^3+\beta^3$ are in G.P., then

(A) $\Delta=0$ (B) $\Delta\neq 0$ (C) $b\Delta=0$ (D) $c\Delta=0$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Physics – questions to revisit

15 questions · easy misses first
cropped from the question paper

Q26 — Vectors

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **91%** got it right · Campus: **86%** (54 of 63)

Easy miss
You: Wrong

26. ABCDEF is regular hexagon with point O as centre. The value of $\overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}+\overrightarrow{AE}+\overrightarrow{AF}$ is

(A) $2\overrightarrow{AB}$ (B) $4\overrightarrow{AO}$ (C) $6\overrightarrow{AO}$ (D) 0

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28 — Vectors – Resultant ⊥ smaller vector; find magnitudes

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **77%** got it right · Campus: **30%** (19 of 63)

Easy miss
You: Wrong

28. The simple sum of two co-initial vectors is 16 units . Their vector sum is 8 units. The resultant of the vectors is perpendicular to the smaller vector . The magnitudes of the two vectors are

(A) 22units and 14 units (B) 4 units and 12 units
(C) 6 units and 10 units (D) 8 units and 12 units

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32

Kinematics

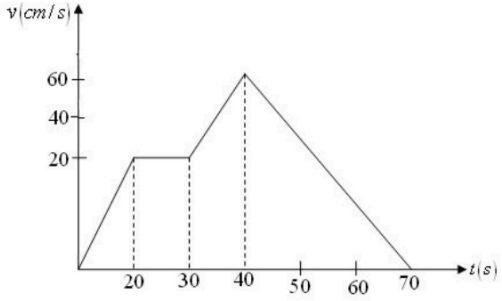
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **75%** got it right · Campus: **29%** (18 of 63)

Easy miss

You: Wrong

32. The velocity versus time curve of a moving points is given below. The maximum acceleration is



(A) $1\ \text{cm/s}^2$ (B) $2\ \text{cm/s}^2$ (C) $3\ \text{cm/s}^2$ (D) $4\ \text{cm/s}^2$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29

Vectors – Angles between three vectors summing to zero

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **62%** got it right · Campus: **24%** (15 of 63)

Within reach

You: Wrong

29. Given: $\vec{a}+\vec{b}+\vec{c}=0$. Out of the three vectors $\vec{a},\vec{b}$ and $\vec{c}$, two are equal in magnitude. The magnitude of the third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. The angles between the vector are

(A) $90^0,135^0,135^0$ (B) $30^0,60^0,90^0$

(C) $45^0,45^0,90^0$ (D) $45^0,60^0,90^0$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q40

Laws of Motion – Velocity over fixed distance vs mass (v ∝ 1/√m)

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **62%** got it right · Campus: **30%** (19 of 63)

Within reach

You: Wrong

40. A constant force acts on a mass m at rest. Velocity acquired in travelling a fixed distance is directly proportional to

(A) $\sqrt{m}$ (B) m (C) $1/\sqrt{m}$ (D) $1/m$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q44 **Laws of Motion – Normal reactions on a cylinder resting on edges**

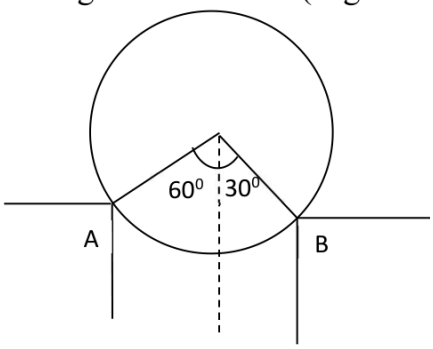
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **53%** got it right · Campus: **29%** (18 of 63)

Within reach

You: Wrong

44. A cylinder of mass M and radius R is resting on two corner edges A and B as shown in the figure. The normal reaction at the edges A and B are (neglect friction)



(A) $N_A = \sqrt{2}N_B$ (B) $N_B = \sqrt{3}N_A$ (C) $N_A = \sqrt{3}N_B$ (D) $N_B = \frac{2\sqrt{3}Mg}{5}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Laws of Motion – Tension / normal reaction in two-block systems (matrix match)

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **42%** got it right · Campus: **30%** (19 of 63)

Within reach

You: Wrong

30. Column-I gives four different situations involving two blocks of mass m_1 and m_2 placed in different ways on a smooth horizontal surface as shown. In each of the situations horizontal forces F_1 and F_2 are applied on blocks of mass m_1 and m_2 respectively and also $m_2 F_1 < m_1 F_2$. Match the statements in column I with corresponding results in column-II

Column-I	Column-II
(A)  . Both the blocks are connected by massless inelastic string. The magnitude of tension in the string is	(p) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_1}{m_1}-\frac{F_2}{m_2}\right)$
(B)  . Both the blocks are connected by massless inelastic string. The magnitude of tension in the string is	(q) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_1}{m_1}+\frac{F_2}{m_2}\right)$
(C)  . The magnitude of normal reaction between the blocks is	(r) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_2}{m_2}-\frac{F_1}{m_1}\right)$
(D)  . The magnitude of normal reaction between the blocks is	(s) $m_1 m_2\left(\frac{F_1+F_2}{m_1+m_2}\right)$

(A) (A) – r ; (B) – q ; (C) – q ; (D) – r

(B) (A) – q ; (B) – r ; (C) – r ; (D) – q

(C) (A) – q ; (B) – r ; (C) – q ; (D) – r

(D) (A) – q ; (B) – p ; (C) – r ; (D) – r

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q35

Kinematics – a = kv² – integrate to get v(s)

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **41%** got it right · Campus: **5%** (3 of 63)

Within reach

You: Skipped

35. If a particle accelerates with $a = kv^2$ and initial velocity = u, then find velocity after displacement s

(A) $v = ue^{ks}$ (B) $v = u^{ks}e^{s}$ (C) $v = u^{k}e^{s}$ (D) $v = eu^{ks}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q38

Kinematics – Resistance ∝ v² – time inside plank

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **27%** got it right · Campus: **11%** (7 of 63)

Tough one

You: Skipped

38. Having gone through a plank of thickness h, a bullet changed its velocity from v_0 to v. Find the time of motion of the bullet in the plank assuming resistance force to be proportional to the square of the velocity

(A) $\left[\frac{h(v_0-v)}{vv_0 l\,n\frac{v_0}{v}}\right]$ (B) $\left[\frac{(v_0-v)}{hvv_0 l\,n\left(\frac{v_0}{v}\right)}\right]$ (C) $\left[\frac{vv_0 \ln\left(\frac{v_0}{v}\right)}{h(v_0-v)}\right]$ (D) $\left[\frac{\ln\frac{v_0}{v}}{vv_0(v_0-v)}\right]$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q45

Laws of Motion – Contact force between pushed blocks (statement type)

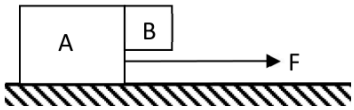
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **27%** got it right · Campus: **10%** (6 of 63)

Tough one

You: Wrong

45. Statement-1: Block A is moving on a horizontal surface towards right under action of force F. all surfaces are smooth. At the instant shown the force exerted by block A on block B is equal to net force on block B.



Statement-2: From Newton's third law, the force exerted by block A on B is equal in magnitude to force exerted by block B on A.

(A) Statement 1 is true, statement 2 is true, statement 2 is correct explanation for statement 1

(B) Statement 1 is true, statement 2 is true, statement 2 is not correct explanation for statement 1

(C) Statement 1 is true, statement 2 is false

(D) Statement 1 is false, statement 2 is true

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q36

Kinematics – a(t) with modulus – area under a-t

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **24%** got it right · Campus: **5%** (3 of 63)

Tough one

You: Wrong

36. Starting from rest a particle moves in a straight line with acceleration $a = \{2 + |t-2|\}\, m/s^2$. Velocity of the particle at the end of 4s will be

(A) 16 m/s (B) 20 m/s (C) 8 m/s (D) 12 m/s

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q43

Laws of Motion – Normal reaction across an inclined contact

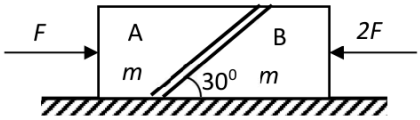
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **20%** got it right · Campus: **10%** (6 of 63)

Tough one

You: Wrong

43. Two blocks 'A" and 'B' each of mass 'm' are placed on a smooth horizontal surface. Two horizontal forces F and 2F are applied on both the blocks 'A' and 'B' respectively as shown in figure. The block A does not slide on block B. Then the normal reaction acting between the two blocks is



(A) F (B) $F/2$ (C) $F/\sqrt{3}$ (D) 3F

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47

Vectors – Net displacement from multi-leg journey

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **15%** got it right · Campus: **3%** (2 of 63)

Tough one

You: Skipped

47. In going from one city to another , a car travels 75 km north, 60 km north –west and 20 km east. The magnitude(in km) of the vector drawn between the two cities is $\left(Take 1/\sqrt{2} = 0.7\right)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q48

Kinematics – Distance from a-t graph

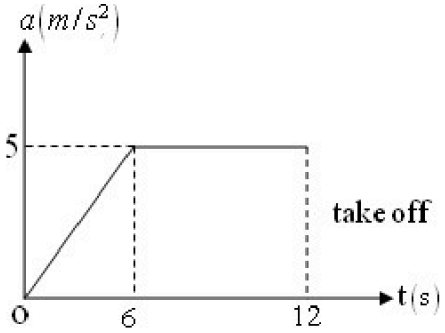
JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **15%** got it right · Campus: **3%** (2 of 63)

Tough one

You: Skipped

48. An experiment on the take off performance of an aeroplane shows that the acceleration varies as shown in the fig and it takes 12 sec to take off from a rest position.The distance along the runway covered by the aeroplane is(in m).



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q49

Kinematics – Free fall then deceleration – find height

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **15%** got it right · Campus: **2%** (1 of 63)

Tough one

You: Skipped

49. A parachutist after bailing out falls 50m without friction. When parachute opens, it decelerates at $2m/\sec^2$. He reaches the ground with a speed of 3m/s. The height he bail out is nearly (in m) (Take g=10m/sec^2) .

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Chemistry – questions to revisit

7 questions · easy misses first
cropped from the question paper

Q61

Atomic Structure
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **84%** got it right · Campus: **40%** (25 of 63)

Easy miss
You: Wrong

61. **Assertion:** Emitted radiation will fall in visible range when an electron jump from n = 4 to n = 2 in H atom.
Reason: Balmer series radiation belongs to visible range for hydrogen atom.
(A)If both (A) and (R) are true, and (R) is the correct explanation of (A).
(B)If both (A) and (R) are true but (R) is not the correct explanation of (A).
(C)If (A) is true but (R) is false.
(D) If (A) is false but (R) is true.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q70

Chemical Bonding
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **72%** got it right · Campus: **52%** (33 of 63)

Easy miss
You: Wrong

70. **Assertion :** Solid NaCl is conducting in nature
Reason : NaCl cantain Na^+, Cl^-.
(A)If both (A) and (R) are true, and (R) is the correct explanation of (A).
(B)If both (A) and (R) are true but (R) is not the correct explanation of (A).
(C)If (A) is true but (R) is false.
(D) If (A) is false but (R) is true.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q74 — Periodic Classification

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5
Top 100: **70%** got it right · Campus: **46%** (29 of 63)

Easy miss
You: Wrong

74. An element X belongs to the 3rd period and has 5 valence electrons. Another element Y belongs to the same period and has 2 valence electrons.
Find the sum of the atomic numbers of X and Y.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q53 — Mole Concept & Basics – Valence electrons in N^{3-} (mole → electrons)

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **65%** got it right · Campus: **25%** (16 of 63)

Within reach
You: Wrong

53. If N_A is Avogadro number, then the number of valence electrons in 4.2 g of N^{3-} ions is
(A) 2.4 N_A (B) 4.2 N_A (C) 1.6 N_A (D) 3.2 N_A

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q75 — Chemical Bonding – Hybrid orbitals + lone pairs in NH_3

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4
Top 100: **60%** got it right · Campus: **30%** (19 of 63)

Within reach
You: Skipped

75. According to VBT, consider the molecule NH_3. The central nitrogen atom undergoes hybridisation before bond formation.
If the number of hybrid orbitals formed is x, and the number of lone pairs present on nitrogen is y, find:
x + y

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q59

Chemical Bonding – VBT, orbital overlap, hybridisation statements

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **50%** got it right · Campus: **29%** (18 of 63)

Within reach

You: Wrong

59. Statement I: According to VBT, a covalent bond is formed by the overlap of half-filled atomic orbitals containing electrons with opposite spins.
Statement II: Greater the extent of effective orbital overlap, stronger is the covalent bond formed.
Statement III: Hybridisation involves the mixing of atomic orbitals of comparable energy to form equivalent hybrid orbitals.
Which of the following is correct?
(A) Only Statement I is correct
(B) Statements I and II are correct
(C) Statements I, II and III are correct
(D) Statements II and III are correct

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q67

Chemical Bonding – Strongest ionic bond (charge and size)

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **32%** got it right · Campus: **13%** (8 of 63)

Tough one

You: Wrong

67. Which combination will give strongest ionic bond?
(A) K^+ and Cl^- (B) K^+ and O^{2-}
(C) Ca^{2+} and Cl^- (D) Ca^{2+} and O^{2-}

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced – question by question

69 / 171 · 40.4%
09 Oct 2026

6 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	6	7	8	9	10	11	12	13	~~14~~	15	16	~~17~~
Physics	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
Chemistry	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	~~50~~	51

7 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 63 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

7 of 15 questions lost · **26** marks · 1 easy miss · 4 within reach · 2 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q3	Easy miss	**Quadratic Equations**	Single correct · +3 / −1	Wrong	4	86%	57%
Q4	Within reach	**Logarithms** – Logs in AP – solve for x	Single correct · +3 / −1	Wrong	4	68%	35%
Q12	Within reach	**Sequences & Series**	Integer · +4 / 0	Wrong	4	68%	38%
Q10	Within reach	**Quadratic Equations** – Roots of two linked quadratics	Integer · +4 / 0	Wrong	4	50%	25%
Q15	Within reach	**Sequences & Series** – AP: ratio of sums of 14 terms	Paragraph · +3 / 0	Wrong	3	42%	8%
Q16	Tough one	**Logarithms** – Log equations – A + B + C	Paragraph · +3 / 0	Wrong	3	18%	3%
Q13	Tough one	**Quadratic Equations** – Spot the polynomial identity, then sum P(1) … P(2023)	Integer · +4 / 0	Wrong	4	– no data*	2%

Physics

14 of 17 questions lost · **56** marks · 3 easy miss · 4 within reach · 7 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q25	Easy miss	**Vectors**	Integer · +4 / 0	Wrong	4	84%	37%
Q19	Easy miss	**Kinematics**	Single correct · +3 / −1	Wrong	4	79%	38%
Q24	Easy miss	**Laws of Motion** – Tensions in two inclined strings	Multiple correct · +4 / −2	Wrong	6	70%	44%
Q21	Within reach	**Kinematics** – Retardation from the t–x relation	Single correct · +3 / −1	Wrong	4	62%	24%
Q20	Within reach	**Kinematics**	Single correct · +3 / −1	Wrong	4	52%	22%
Q23	Within reach	**Laws of Motion** – Block on block, no friction – motion of m	Multiple correct · +4 / −2	Wrong	6	51%	22%
Q30	Within reach	**Kinematics** – a from v(t) at a given position	Integer · +4 / 0	Skipped	4	43%	8%
Q28	Tough one	**Laws of Motion** – Spring-connected blocks – acceleration	Integer · +4 / 0	Wrong	4	25%	13%
Q33	Tough one	**Kinematics** – Train passage – retardation to avoid collision	Paragraph · +3 / 0	Wrong	3	15%	8%
Q34	Tough one	**Kinematics** – Train passage – time of retardation	Paragraph · +3 / 0	Wrong	3	15%	10%
Q27	Tough one	**Laws of Motion** – Normal reaction between pushed blocks	Integer · +4 / 0	Wrong	4	11%	2%
Q31	Tough one	**Vectors** – Carrom passage – displacement centre → edge	Paragraph · +3 / 0	Wrong	3	7%	2%
Q29	Tough one	**Kinematics**	Integer · +4 / 0	Wrong	4	– no data*	6%
Q32	Tough one	**Vectors** – Carrom passage – displacement edge → hole	Paragraph · +3 / 0	Wrong	3	6%	2%

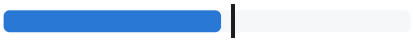
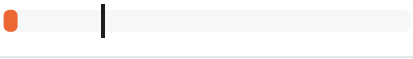
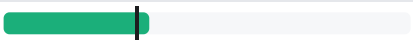
Chemistry

5 of 16 questions lost · **20** marks · 3 easy miss · 1 within reach · 1 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q38	Easy miss	**Chemical Bonding** – Pair of species with the same shape (VSEPR)	Single correct · +3 / −1	Skipped	3	92%	49%
Q40	Easy miss	**Atomic Structure**	Multiple correct · +4 / −2	Wrong	6	82%	44%
Q47	Easy miss	**Mole Concept & Basics** – Mass of water from 8 g H_2 + 32 g O_2 (limiting reagent)	Integer · +4 / 0	Wrong	4	72%	32%
Q35	Within reach	**Mole Concept & Basics** – Limiting reagent over two-step reaction	Single correct · +3 / −1	Wrong	4	51%	24%
Q51	Tough one	**Atomic Structure** – Hydrogen-like ion – find Z from photon energies	Paragraph · +3 / 0	Wrong	3	0%	5%

8 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject. Chapter marks are approximate (≈): partly-correct answers cannot be traced to a question.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Logarithms	4	≈ **6**/13		50%	**Q4**, **Q16**
Sequences & Series	4	≈ **8**/15		56%	**Q12**, **Q15**
Quadratic Equations	7	≈ **14**/26		43%	**Q3**, **Q10**, **Q13**
Physics					
Laws of Motion	4	≈ **-4**/16		6%	**Q23**, **Q24**, **Q27**, **Q28**
Kinematics	8	≈ **1**/27		24%	**Q19**, **Q20**, **Q21**, **Q29**, **Q30**, **Q33**, **Q34**
Vectors	5	≈ **7**/17		23%	**Q25**, **Q31**, **Q32**
Chemistry					
Chemical Bonding	1	≈ **0**/3		41%	**Q38**
Atomic Structure	4	≈ **5**/14		33%	**Q40**, **Q51**
Mole Concept & Basics	9	≈ **25**/33		44%	**Q35**, **Q47**
Periodic Classification	2	≈ **7**/7		68%	– none

Bar = your % of the chapter's marks | campus average

9 JEE Advanced · Maths – questions to revisit

7 questions · easy misses first
cropped from the question paper

Q3 Quadratic Equations

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 4
Top 100: **86%** got it right · Campus: **57%** (36 of 63)

Easy miss
You: Wrong

3. If α, β are roots of the equation $ax^2+bx+c=0$ then the value of $\left(\frac{\alpha}{a\beta+b}\right)^3-\left(\frac{\beta}{a\alpha+b}\right)^3$ is equal to:

A) 0 B) ab C) $b-c$ D) $c-a$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q4 Logarithms – Logs in AP – solve for x

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 4
Top 100: **68%** got it right · Campus: **35%** (22 of 63)

Within reach
You: Wrong

4. If $\log_3 2, \log_3\left(2^x-5\right)$ & $\log_3\left(2^x-\frac{7}{2}\right)$ are in AP then the value of x is?

A) 2 B) 3 C) 4 D) 2&3

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q12 Sequences & Series

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4
Top 100: **68%** got it right · Campus: **38%** (24 of 63)

Within reach
You: Wrong

12. Consider a sequence whose sum to n terms is given by the quadratic function $S_n=3n^2+5n$. For the given sequence, the number 5456 is m^{th} term then value of m is___?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q10

Quadratic Equations – Roots of two linked quadratics

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **50%** got it right · Campus: **25%** (16 of 63)

Within reach

You: Wrong

10. Let a and b be roots of the equation $x^2-10cx-11d=0$ and the roots of $x^2-10ax-11b=0$ are c and d; then the value of $\frac{a+b+c+d}{605}$ is____?(Given $a\neq b\neq c\neq d\neq 0$)

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q15

Sequences & Series – AP: ratio of sums of 14 terms

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **42%** got it right · Campus: **8%** (5 of 63)

Within reach

You: Wrong

Sum of n terms of two different Arithmetic progressions are in the ratio 5n+4:9n+6 then

15. The ratio of the sum of their first 14 terms is Q, then [100Q] is _____(where [.] is G.I.F)

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q16

Logarithms – Log equations – A + B + C

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **18%** got it right · Campus: **3%** (2 of 63)

Tough one

You: Wrong

PARAGRAPH-II

Let **A** denote the product xyz where x, y, z are solutions of the system of equations

$\log_3 x=\log_3 5-\log_3 7, \log_5 y=\log_5 7-\log_5 3, \log_7 z=\log_7 3-\log_7 5$

Let **B** denote the sum of square of the solutions of the equation

$\log_2\left(\log_2 x^6-3\right)-\log_2\left(\log_2 x^4-5\right)=\log_2 3$

Let **C** denote the characteristic of the number K, where

$K=\log_2\left(\log_2 3\right)-\log_2\left(\log_4 3\right)+\log_2\left(\log_4 5\right)-\log_2\left(\log_6 5\right)+\log_2\left(\log_6 7\right)-\log_2\left(\log_8 7\right)$

16. The value of A+B+C is____?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q13 **Quadratic Equations – Spot the polynomial identity, then sum P(1) ... P(2023)**

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **no data** got it right · Campus: **2%** (1 of 63)

Tough one

You: Wrong

13. Let $P(x)=(1+\sqrt{2})(1+\sqrt{3})(x-\sqrt{2})(x-\sqrt{3})+(1+\sqrt{3})(\sqrt{3}+\sqrt{2})(x-1)(x-\sqrt{2})$ $-2(1+\sqrt{2})(\sqrt{3}+\sqrt{2})(x-1)(x-\sqrt{3})\ \forall x\in R$. Then the value of $P(1)+P(2)+P(3)+...P(2023)$ is_____?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Physics – questions to revisit

14 questions · easy misses first
cropped from the question paper

Q25

Vectors
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **84%** got it right · Campus: **37%** (23 of 63)

Easy miss
You: Wrong

25. The vectors $\vec{A} = 5\hat{i} - 7\hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} + 2\hat{j} - c\hat{j}$ are perpendicular. Find the value of C.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q19

Kinematics
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4
Top 100: **79%** got it right · Campus: **38%** (24 of 63)

Easy miss
You: Wrong

19. A juggler throws balls into air. He throws one whenever the previous one is at its highest point. How high do the balls rise if he throws n balls each second? Acceleration due to gravity is g.

A) g/n^2 B) $g/2n^2$ C) $2gn^2$ D) $2n^2/g$

Why did I lose this question? (write the reason and the correct approach)

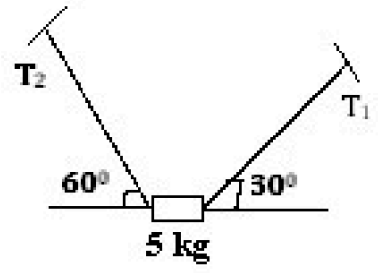
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q24

Laws of Motion – **Tensions in two inclined strings**
JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **70%** got it right · Campus: **44%** (28 of 63)

Easy miss
You: Wrong

24. A body of mass 5 kg is suspended by the strings making angles 60^0 and 30^0 with the horizontal as shown in the figure($g = 10ms^{-2}$). Then



A) $T_1 = 25N$ B) $T_2 = 25N$ C) $T_1 = 25\sqrt{3}N$ D) $T_2 = 25\sqrt{3}N$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q21

Kinematics – Retardation from the t-x relation

JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **62%** got it right · Campus: **24%** (15 of 63)

Within reach

You: Wrong

21. What is the retardation of a moving particle if the relation between time (t) and position (x) is $t = Ax^2 + Bx$, where A and B are appropriate constants.

A) $2A/(2Ax+B)^3$

B) $2B/(2Ax+B)^3$

C) $-2A/(2Ax+B)^3$

D) $-2B/(2Ax+B)^3$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20

Kinematics

JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **52%** got it right · Campus: **22%** (14 of 63)

Within reach

You: Wrong

20. A particle is projected vertically upwards from the ground at time $t = 0$ and reaches a height h at $t = T$. Greatest height of the particle is

A) $(gT^2+h)^2/8gT^2$

B) $2(gT^2+h)^2/8gT^2$

C) $(gT^2+2h)^2/8gT^2$

D) $2h$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q23

Laws of Motion – Block on block, no friction – motion of m

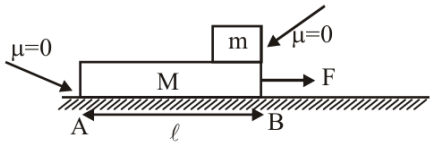
JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 6

Top 100: **51%** got it right · Campus: **22%** (14 of 63)

Within reach

You: Wrong

23. In the figure small block is kept on M, then



A) the acceleration of m w.r.t. ground is $\frac{F}{m}$

B) the acceleration of m w.r.t. ground is zero

C) the time taken by m to separate from M is $\sqrt{\frac{2\ell m}{F}}$

D) the time taken by m to separate from M is $\sqrt{\frac{2\ell M}{F}}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Kinematics – a from v(t) at a given position

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **43%** got it right · Campus: **8%** (5 of 63)

Within reach

You: Skipped

30. A body starts from origin and moves along x axis such that at any instant velocity is $v_t=4t^3-2t$ where t is in second and v_t is in ms^{-1}. The acceleration of the particle when it is 2m from the origin is $(20+K)\,ms^{-2}$, where K is

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28

Laws of Motion – Spring-connected blocks – acceleration

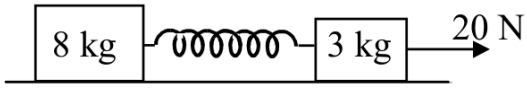
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **25%** got it right · Campus: **13%** (8 of 63)

Tough one

You: Wrong

28. If the acceleration of 3 kg block is 4 m/s^2 to the left. What is the magnitude of acceleration of 8 kg block.



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q33

Kinematics – Train passage – retardation to avoid collision

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **15%** got it right · Campus: **8%** (5 of 63)

Tough one

You: Wrong

PASSAGE-II:

The driver of a train A moving with a uniform speed of 144 km/h sights another train B, 1km ahead of him. The train B is moving with a uniform speed of 108 km/h. the driver of the train A immediately applies brakes producing a constant retardation and just manages to avoid a collision.

33. What is the retardation of the train A(in ms^{-2})_____

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34

Kinematics – Train passage – time of retardation

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **15%** got it right · Campus: **10%** (6 of 63)

Tough one

You: Wrong

PASSAGE-II:

The driver of a train A moving with a uniform speed of 144 km/h sights another train B, 1km ahead of him. The train B is moving with a uniform speed of 108 km/h. the driver of the train A immediately applies brakes producing a constant retardation and just manages to avoid a collision.

34. For how long is this retardation produced(in sec)_____

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Laws of Motion – Normal reaction between pushed blocks

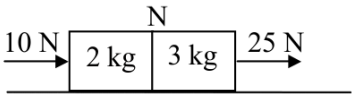
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **11%** got it right · Campus: **2%** (1 of 63)

Tough one

You: Wrong

27. The normal reaction between 2 kg and 3 kg block is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q31

Vectors – Carrom passage – displacement centre → edge

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **7%** got it right · Campus: **2%** (1 of 63)

Tough one

You: Wrong

PASSAGE-1:

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line.

Answer the following question.

31. The magnitude of the displacement of the queen from the centre to the front edge is $\frac{2}{3}\sqrt{a}\,ft$, the value of a is _____

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29

Kinematics

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **no data** got it right · Campus: **6%** (4 of 63)

Tough one

You: Wrong

29. A lead ball is dropped freely into a lake from a diving board 5 m above the water. It strikes the water with a certain velocity and then reaches the bottom with the same constant velocity in 5 second after it is dropped.

Take g=10 ms^{-2} calculate the average velocity of the ball in ms^{-1} just before reaches to the bottom from the freely dropped point .

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32 **Vectors – Carrom passage – displacement edge → hole**

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **6%** got it right · Campus: **2%** (1 of 63)

Tough one

You: Wrong

PASSAGE-1:

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line. Answer the following question.

32. The magnitude of the displacement of the queen from the front edge to the hole is $\frac{4}{3}\sqrt{b}\ ft$, the value of b is_____

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Chemistry — questions to revisit

5 questions · easy misses first
cropped from the question paper

Q38 — Chemical Bonding — Pair of species with the same shape (VSEPR)

JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 3
Top 100: **92%** got it right · Campus: **49%** (31 of 63)

Easy miss
You: Skipped

38. The pair of species with similar shape is

(A) PCl_3, NH_3 (B) CF_4, SF_4 (C) $PbCl_2$, CO_2 (D) PF_5, IF_5

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q40 — Atomic Structure

JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **82%** got it right · Campus: **44%** (28 of 63)

Easy miss
You: Wrong

40. Which of the following properties is/are proportional to the energy of the electromagnetic radiation ?

(A) Frequency (B) Wave number
(C) Wavelength (D) Number of photons

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47 — Mole Concept & Basics — Mass of water from 8 g H_2 + 32 g O_2 (limiting reagent)

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4
Top 100: **72%** got it right · Campus: **32%** (20 of 63)

Easy miss
You: Wrong

47. For $2H_2 + O_2 \rightarrow 2H_2O$, 8 g H_2 reacts with 32 g O_2. Find the mass of water formed.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q35 — Mole Concept & Basics – Limiting reagent over two-step reaction

JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4

Top 100: **51%** got it right · Campus: **24%** (15 of 63)

Within reach

You: Wrong

35. 5 moles of A, 6 moles of Z and mixed with sufficient amount of C to produce final product F. How many maximum moles of 'F' can be produced as per the given sequence of reaction ?

$A + 2Z \rightarrow B$; $B + C \rightarrow Z + F$

(A) 3 (B) 2 (C) 4 (D) 5

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q51 — Atomic Structure – Hydrogen-like ion – find Z from photon energies

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **0%** got it right · Campus: **5%** (3 of 63)

Tough one

You: Wrong

PARAGRAPH-II

When electron jumps from lower orbit to higher orbit energy is gained and when electron jumps from higher orbit to lower orbit energy is released.

51. A hydrogen like species (atomic number Z) is present in a higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successive emission of two photons of energies 10.20 eV and 17.0 eV respectively. Alternatively the atom from the same excited state can make a transition to the second excited state by successive emission of two photons of energy 4.25 eV and 5.95 eV respectively. Determine the value of Z.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

10 How to read this report

Where the numbers come from. Marks, percentages, All-India / State / Campus ranks, subject marks, subject All-India ranks and the wrong / skipped questions are the official results in the institute's *All India Marks Analysis* (graded with the final answer key). Maximum marks leave out deleted questions — JEE Mains: Q20 — JEE Advanced: Q14, Q17, Q50. **Program rank** = rank in Pragya G8 (EXI) Achievers across all campuses, from the batch's question-wise sheet († JEE Advanced: that sheet was graded before the final key, so this rank may differ slightly). **Top 100** = the batch's top 100 across all campuses, from the same question-wise sheet; * where it has no data for a question, the campus figure decides the priority. In JEE Advanced that sheet predates the final key, so its figures for the revised questions Q25, Q48, Q51 use the old answers. **Campus** = all students of this program on campus. **Priority:** Easy miss = 70%+ of the top 100 got it right; Within reach = 40–69%; Tough one = below 40%. **Marks lost** on a question = its marks plus the negative mark if answered wrong, its marks if skipped. **Weekly-test average** = your average over the weekly Mains tests (WTM) or Advanced tests (WTA) before this Part Test, from the campus averages sheet. The weekly Advanced tests are marked out of 240; their averages are scaled to 180 to match this paper (× 180/240, so 80 per subject becomes 60). Percentages are unchanged.